

JEE MAIN REVISION MASTER: CIRCLE 2026 1

QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Circle - Core Application	3 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Circle - Core Application (Moderate Yield — 3 PYQs)

Core Principle: Key problem pattern in Kcet Circle.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area enclosed by the curve $x=\sqrt{3}\cos\theta, y=\sqrt{3}\sin\theta$ is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "If in two circles, arcs of the same length subtend angles 30° and 78° at the centre, then the ratio of their radii is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The maximum area of a rectangle inscribed in the circle $(x+1)^2+(y-3)^2=64$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area enclosed by the curve $x=\sqrt{3}\cos\theta, y=\sqrt{3}\sin\theta$ is

- (A) $\sqrt{3}\pi$ sq. units
- (B) 9π sq. units
- (C) 6π sq. units
- (D) 3π sq. units

Correct Answer: None

Solution Analysis: Given, $x=\sqrt{3}\cos\theta, y=\sqrt{3}\sin\theta$ We know the standard parametric form of a circle is $x=a\cos\theta, y=a\sin\theta$. This represents a circle with centre at the origin and radius a . Here, $a=\sqrt{3}$. So the radius of the circle is $\sqrt{3}$. Area enclosed by a circle is πr^2 . So, $\text{Area}=\pi(\sqrt{3})^2$. $\text{Area}=3\pi$. Hence, the area enclosed by the curve is 3π sq. units. Correct option: Option D.